

Departement Elektriese, Elektroniese en
Rekenaar-Ingenieurswese

Semestertoets 2

VRAESTEL

Kopiereg voorbehou

Module EBN111
Elektrisiteit en Elektronika
Mei 2018

Department of Electrical, Electronic and
Computer Engineering

Semester Test 2 QUESTION PAPER

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Module EBN111
Electricity and Electronics
May 2018



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Eksamensinligting:

Exam information:

Maksimum punte: <i>Maximum marks:</i>	46	Volpunte: <i>Full marks:</i>	45
Duur van vraestel: <i>Duration of paper:</i>	90 min + 10 min to complete OCR form <i>90 min + 10 min om OKH vorm te voltooi</i>	Oopboek / toeboek: <i>Open / closed book:</i>	Toe <i>Closed</i>
Totale aantal bladsye (hierdie blad ingesluit): <i>Total number of pages (including this page):</i>		15 + 1 Answer sheets	

BELANGRIK- IMPORTANT

- Die eksamenregulasies van die Universiteit van Pretoria geld.
The examination regulations of the University of Pretoria apply.
- Alle vrae moet op die ANTWOORDBLAD en op die Optiese Karakter Herkenning (OKH) vorm ingevul word.**
All questions must be answered on the ANSWER SHEET and on the Optical Character Recognition (OCR) form.
- Neem noukeurig kennis van die instruksies op die OKH vorm.
Take careful consideration of the instructions on the OCR form.
- Antwoorde moet eenhede insluit!
Answers must include units!
- Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.**
Final answers must be written as real numbers, and not as fractions.

Dosent(e): <i>Lecturer(s):</i>	1. D le Roux
	2. D Ramotsoela
	3. X Ye

STEPS TO FOLLOW

Do step 0 before the start of the session.

Do step 1 in the first 90 minutes of the test.

Do step 2 within the following 10 minutes.

STEP 0 (-5 min to 0 min): Complete the front page of the OCR form.

STEP 1 (0 min to 90 min):

- Do your own calculations on the QUESTION PAPER. The question paper **will not** be taken in at the end of the test.
- Write down the answers for each question on the PRACTICE ANSWER SHEET at the end of the QUESTION PAPER. The PRACTICE ANSWER SHEET **will not** be taken in at the end of the test.

STEP 2 (90 min to 100 min):

Carefully copy the answers from the PRACTICE ANSWER SHEET to the OCR form.

- The Assessment ID is: **2018EBN111S2**
- Include the correct unit for each answer. Apply appropriate prefixes.
- Use a mechanical pencil with 0.5mm HB lead to complete the form.
- Each cell should only have one character, and characters may not touch or cross the cell.
- Do not use commas! Please make use of full stops.

- Complex numbers:

Rectangular form: always place the “j” **at the start** of the imaginary part of the complex number.

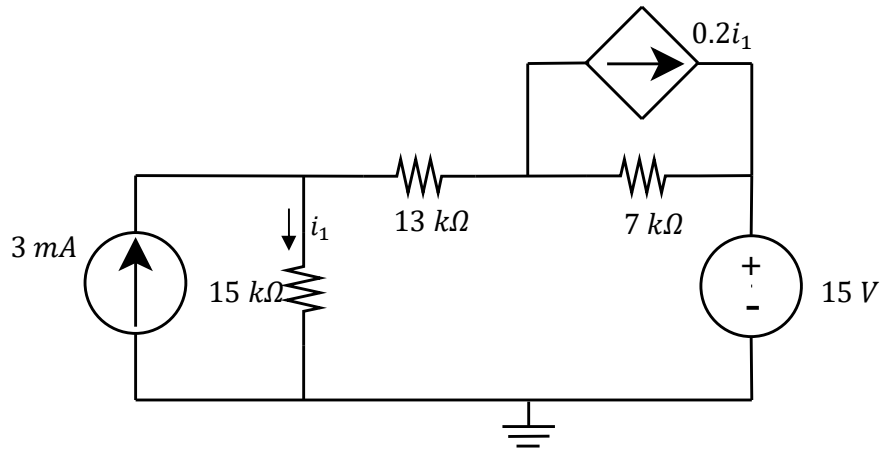
Polar form: always add the “°” (degree sign) after the angle, and make a clear “∠” (angle sign).

Examples

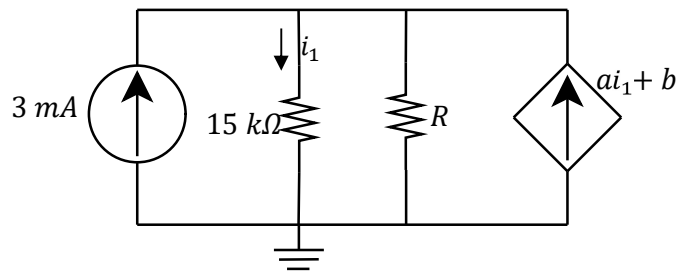
01	–	1	5	4	.	9				k	Ω							
02	1	.	2							r	a	d	÷	s				
03	–	1	.	4	+	j	8			μ	A							
04	1	4	0	°														
05	3	4	0	∠	4	5	°			V								
06	–	j	5	4						k	Ω							
07	1	∠	2	0	°													
08	5	0	∠	–	1	3	0	°		m	A							

SECTION / AFDELING 1 [12]

Questions 1-3 [4]		Vraag 1-3 [4]	
Simplify the circuit (a) with source transformation so that it looks like circuit (b) and determine:		Vereenvoudig kringbaan (a) m.b.v. brontransformasie sodat dit lyk soos kringbaan (b), en bepaal:	
01	R	01	R
02	a	02	a
03	b	03	b



(a)



(b)

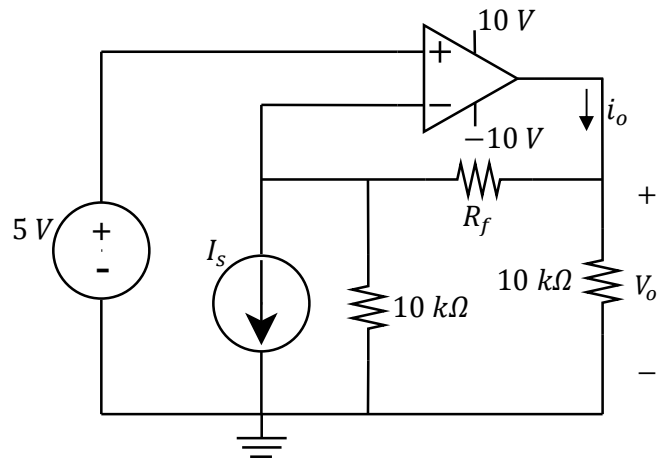
Questions 4-5 [4]		Vraag 4-5 [4]	
Determine the Thevenin equivalent voltage V_{TH} and resistance R_{TH} w.r.t. terminals a - b .		Bepaal die Thevenin ekwivalente spanning V_{TH} en weerstand R_{TH} met verwysing tot terminale a en b .	
04	V_{TH}	04	V_{TH}
05	R_{TH}	05	R_{TH}

Question 6-7 [4]		Vraag 6-7 [4]	
Determine the Norton equivalent current I_N and resistance R_N w.r.t. terminals a - b .		Bepaal die Norton ekwivalente stroom I_N en weerstand R_N met verwysing tot terminale a en b .	
06	I_N	06	I_N
07	R_N	07	R_N

The circuit diagram shows a 15V DC voltage source with its positive terminal at the top. It is connected in series with a $12\text{ k}\Omega$ resistor. This series combination is connected to a node that branches into two parallel paths. The first path goes down through a $6\text{ k}\Omega$ resistor to the common ground. The second path goes through terminal a , which is connected to the node between the $12\text{ k}\Omega$ and $6\text{ k}\Omega$ resistors. Terminal b is connected to the node between the $6\text{ k}\Omega$ resistor and the ground. The other end of the $12\text{ k}\Omega$ resistor is connected to the top rail, which is also connected to the positive terminal of the voltage source. The bottom rail is connected to the negative terminal of the voltage source and the ground symbol.

SECTION/AFDELING 2 [16]

Question 8-9 [4]		Vraag 8-9 [4]	
08	What is the maximum value of R_f before the operational amplifier reaches saturation, given $I_s = 0\text{ A}$?	08	Gegee $I_s = 0\text{ A}$, wat is die maksimum waarde van R_f voordat die operasionele versterker versadiging bereik?
09	Calculate V_o if $I_s = 2\text{ mA}$ and $R_f = 1\text{ k}\Omega$.	09	Bereik V_o indien $I_s = 2\text{ mA}$ en $R_f = 1\text{ k}\Omega$.



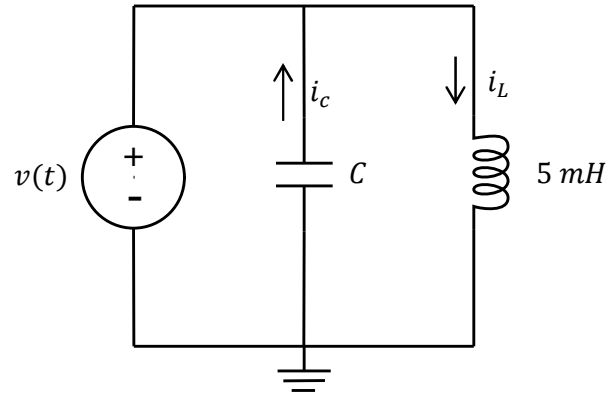
Questions 10-11 [4]		Vraag 10-11 [4]	
10	Find v_s if $I_1 = 0\text{ A}$ and $v_l = 12\text{ V}$.	10	Bepaal v_s indien $I_1 = 0\text{ A}$ en $v_l = 12\text{ V}$.
11	Find v_l if $I_1 = 5\text{ mA}$ and $v_s = 0\text{ V}$.	11	Bepaal v_l indien $I_1 = 5\text{ mA}$ en $v_s = 0\text{ V}$.

Question 12-13 [4]		Vraag 12-13 [4]	
12	Find v_2 if $v_3 = 5\text{ V}$ and $v_I = 0\text{ V}$.	12	Bepaal v_2 indien $v_3 = 5\text{ V}$ en $v_I = 0\text{ V}$.
13	Find v_3 if $v_2 = 0\text{ V}$ and $v_I = 9\text{ V}$.	13	Bepaal v_3 indien $v_2 = 0\text{ V}$ en $v_I = 9\text{ V}$.

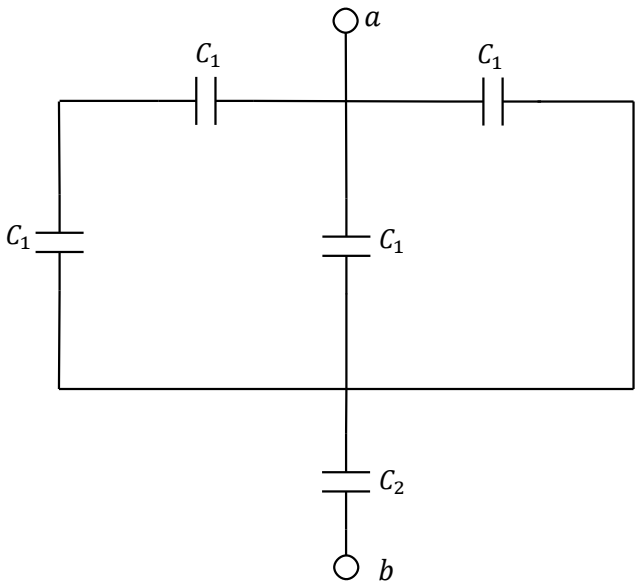
Questions 14-15 [4]		Vraag 14-15 [4]	
Find R_1 and R_2 if $V_0 = -13\text{ V}$ and $V_X = 18\text{ V}$.		Bepaal R_1 en R_2 indien $V_0 = -13\text{ V}$ en $V_X = 18\text{ V}$.	
14	R_1	14	R_1
15	R_2	15	R_2

SECTION/AFDELING 3 [10]

Question 16-17 [4]		Vraag 16-17 [4]	
16	Find $v(1ms)$ if $i_L(t) = 5 + 3e^{-2t} mA$	16	Bepaal $v(1ms)$ indien $i_L(t) = 5 + 3e^{-2t} mA$.
17	Find C if $i_C(10s) = -5 A$ and $v(t) = 5t^2 - 20t V$.	17	Bepaal C indien $i_C(10s) = -5 A$ en $v(t) = 5t^2 - 20t V$.



Question 18-19 [4]		Vraag 18-19 [4]	
Assume steady-state conditions (direct-current) for the circuit below.		Aanvaar gestadigde toestand (gelykstroom) vir onderstaande kringbaan.	
18	Find V_S if the energy stored in the capacitor is 150 mJ.	18	Bepaal V_S indien die energie gestoor in die kapasitor gelyk is aan 150 mJ.
19	Find V_I if the energy stored in the inductor is 2 μ J.	19	Bepaal V_I indien die energie gestoor in die induktor gelyk is aan 2 μ J.

Question 20 [2]		Vraag 20 [2]	
Determine the equivalent capacitance between terminals a-b if $C_1 = 120\mu\text{F}$ and $C_2 = 150\mu\text{F}$.		Bepaal die ekwivalente kapasitansie tussen terminale a-b indien $C_1 = 120\mu\text{F}$ en $C_2 = 150\mu\text{F}$	
20	C_{ab}	20	C_{ab}
			

SECTION / AFDELING 3 [8]

Question 21-22 [4]		Vraag 21-22 [4]	
21	Simplify the following expression of $\vec{V}$ and write the answer in polar form: $\vec{V} = \frac{(\sqrt{8} + j\sqrt{8})^2}{\sqrt[3]{-j8} + 4.685\angle 92.07^\circ}$	21	Vereenvoudig die volgende uitdrukking vir $\vec{V}$ en skryf die antwoord in polêre vorm: $\vec{V} = \frac{(\sqrt{8} + j\sqrt{8})^2}{\sqrt[3]{-j8} + 4.685\angle 92.07^\circ}$
22	The following sinusoid is given: $v(t) = 20 \cos(\omega t + 60^\circ) \text{ mV}$ Find $v(10\text{ms})$ if the frequency is $f = 50 \text{ Hz}$.	22	Die volgende sinusgolf word gegee: $v(t) = 20 \cos(\omega t + 60^\circ) \text{ mV}$ Bepaal $v(10\text{ms})$ indien die frekwensie $f = 50 \text{ Hz}$ is.

Questions 23-25 [4]		Vraag 23-25 [4]	
Given two sinusoid signals v_1 and v_2 where $v_1 = 2 \cos(10t - 123^\circ),$ $v_2 = -2.015 \sin(10t - 23.04^\circ).$		Twee sinusgolwe v_1 en v_2 word gegee: $v_1 = 2 \cos(10t - 123^\circ),$ $v_2 = -2.015 \sin(10t - 23.04^\circ).$	
23	Determine $v = v_1 - v_2$ and write the final answer in <i>polar form</i> .	23	Bepaal $v = v_1 - v_2$ en skryf die finale antwoord in <i>polêre vorm</i> .
24	Which statement is correct (write either 1 or 2)? 1. v_1 lags v_2 2. v_1 leads v_2	24	Watter stelling is korrek (skryf of 1 of 2)? 1. v_1 loop na v_2 2. v_1 loop voor v_2
25	What is the phase angle ϕ_d between v_1 and v_2 ? (Note: $0 \leq \phi_d \leq 180^\circ$)	25	Wat is die fashoek ϕ_d tussen v_1 en v_2 ? (Let op: $0 \leq \phi_d \leq 180^\circ$)

PRACTICE PAGE / OEFENBLAD

Assessment ID

2	0	1	8	E	B	N	1	1	1	S	2		
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Answer

Unit

01	$R =$	
02	$a =$	(No unit)
03	$b =$	(No unit)
04	$V_{TH} =$	
05	$R_{TH} =$	
06	$I_N =$	
07	$R_N =$	
08	$R_f =$	
09	$V_o =$	
10	$v_s =$	
11	$v_l =$	
12	$v_2 =$	
13	$v_3 =$	
14	$R_l =$	
15	$R_2 =$	
16	$v(1ms) =$	
17	$C =$	
18	$V_S =$	
19	$V_l =$	
20	C_{ab}	
21	$\vec{v} =$	
22	$v(10ms) =$	
23	$\vec{v} =$	
24	Correct statement =	(No unit)
25	$\emptyset_d =$	